HPDP PROJECT 2 SECTION 02

SHOPEE APPLICATION REAL-TIME SENTIMENT ANALYSIS USING APACHE SPARK AND KAFKA

GROUP BigPotato

PROBLEM

- Massive volume of Google Play reviews for Shopee
- Manual analysis is inefficient
- Need timely customer insights

SOLUTION

- NLP for text preprocessing
- Machine learning for sentiment classification
- Real-time analytics using Kafka & Spark

PROJECT OBJECTIVES

- Collect Shopee Malaysia Google Play reviews
- Perform NLP preprocessing
- Train & compare sentiment models
- Build Kafka–Spark streaming pipeline
- Visualize results using Elasticsearch result
- Compare Batch vs Streaming performance

DATA COLLECTION

COLLECTION PROCESS

Data Source

- Public customer reviews and ratings
- Real-world Malaysian customer feedback

Crawling Engine

Google Play Scraper (Python)

Package Verification (com.shopee.my) → **Pagination** → **Metadata Extraction**

Output

- 10000 rows records
- 6 attributes
- CSV file

RAW DATA

No	Attribute	Description	Example
1	Review ID	Unique review identifier	12c34620-7baa-4041-85a9-2fc1ff43a864
2	Rating	1–5 stars	5
3	Review Date	Submission date	6/28/2026
4	App Version	App version	3.56.15
5	Thumbs Up	Helpful votes	5
6	Original Text	Customer review	"Fast delivery and many vouchers."

DATA PROCESSING

DATA CLEANING

- Removed duplicate review IDs and duplicate review texts
- Removed missing, empty, and low-information reviews (less than 3 words)
- Reduced dataset from 10,000 to 6,395 cleaned reviews

NLP PREPROCESSING

- Converted all text to lowercase
- Text cleaning (URL, punctuation, emoji, numbers)
- Tokenized review text into individual words
- Removed English and Malay stop words
- Applied English lemmatization
- Generated a new cleaned_text column while preserving the original review

RAW DATA

10000 rows
6 columns

CLEANED DATA

6395 rows
7 columns

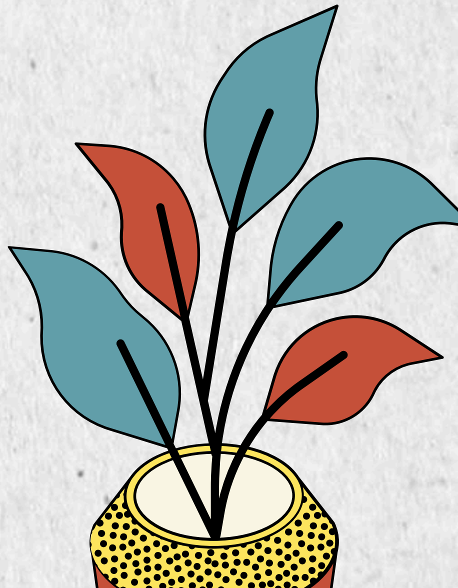
CLEANED DATA

No	Attribute	Description	Example
1	Review ID	Unique review identifier	12c34620-7baa-4041-85a9-2fc1ff43a864
2	Rating	1–5 stars	5
3	Review Date	Submission date	6/28/2026
4	App Version	App version	3.56.15
5	Thumbs Up	Helpful votes	5
6	Original Text	Customer review	shopee very nice and very easy
7	Cleaned Text	Review after NLP preprocessing	shopee nice easy

SENTIMENT MODEL DEVELOPMENT

MODEL DEVELOPMENT

- **Naive Bayes** — TF-IDF + Laplace smoothing (alpha=0.1)
- **Logistic Regression** — TF-IDF + balanced class weighting (C=0.5)
- **Linear SVM** — TF-IDF + balanced class weighting (C=0.5)



MODEL COMPARISON

Model	Accuracy	Weighted F1	Macro F1
Naive Bayes ⭐	**88.19%**	**0.8663**	0.59
Logistic Regression	84.75%	0.8475	0.63
Linear SVM	86.40%	0.8592	**0.64**

📌 **Best Model Selected: Multinomial Naive Bayes**
Reason: Achieved the highest Weighted F1-Score (0.8663) and Accuracy (88.19%) while ensuring high computational efficiency for the real-time Spark streaming pipeline.

APACHE SYSTEM ARCHITECTURE

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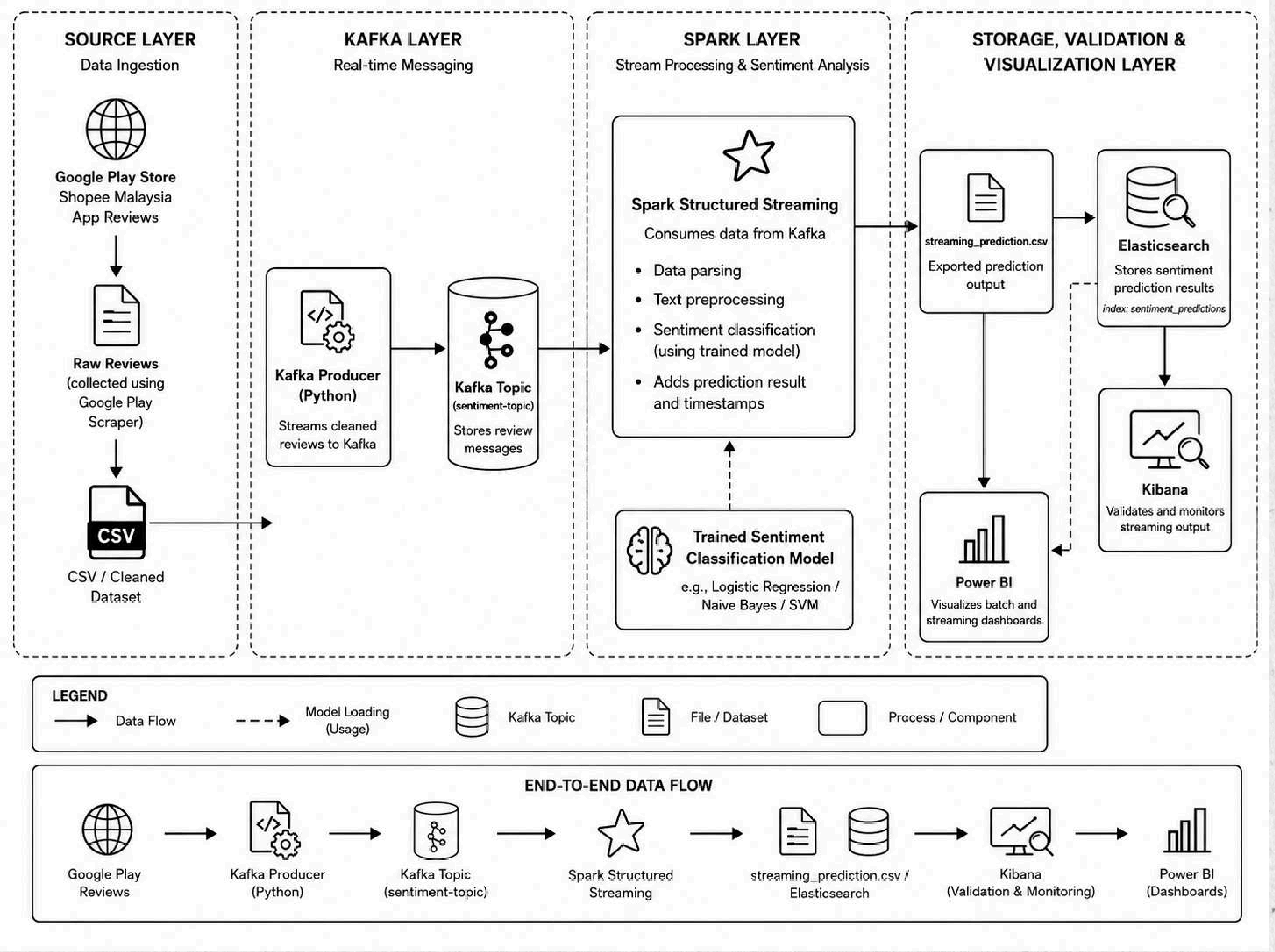


The real-time pipeline connects Kafka, Spark Structured Streaming, Elasticsearch, Kibana, and Power BI to process Shopee review sentiment continuously.

REAL-TIME PIPELINE IMPLEMENTATION

1. Kafka Data Ingestion

- Reads cleaned review data row by row
- Converts each review into JSON message
- Sends messages to sentiment-topic

2. Spark Structured Streaming Processing

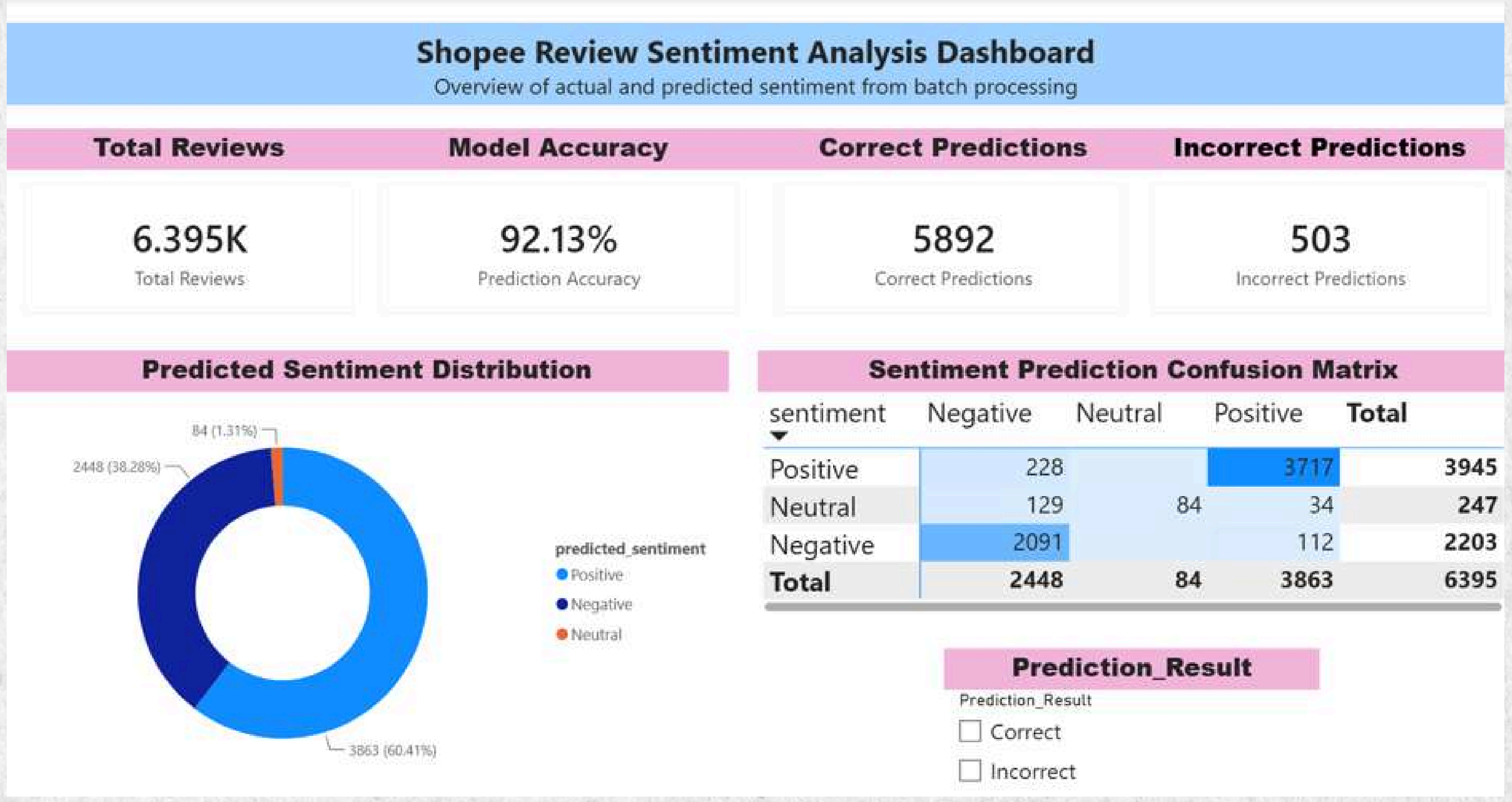
- Consumes messages from Kafka
- Applies trained sentiment model to cleaned_text
- Processes data in micro-batches

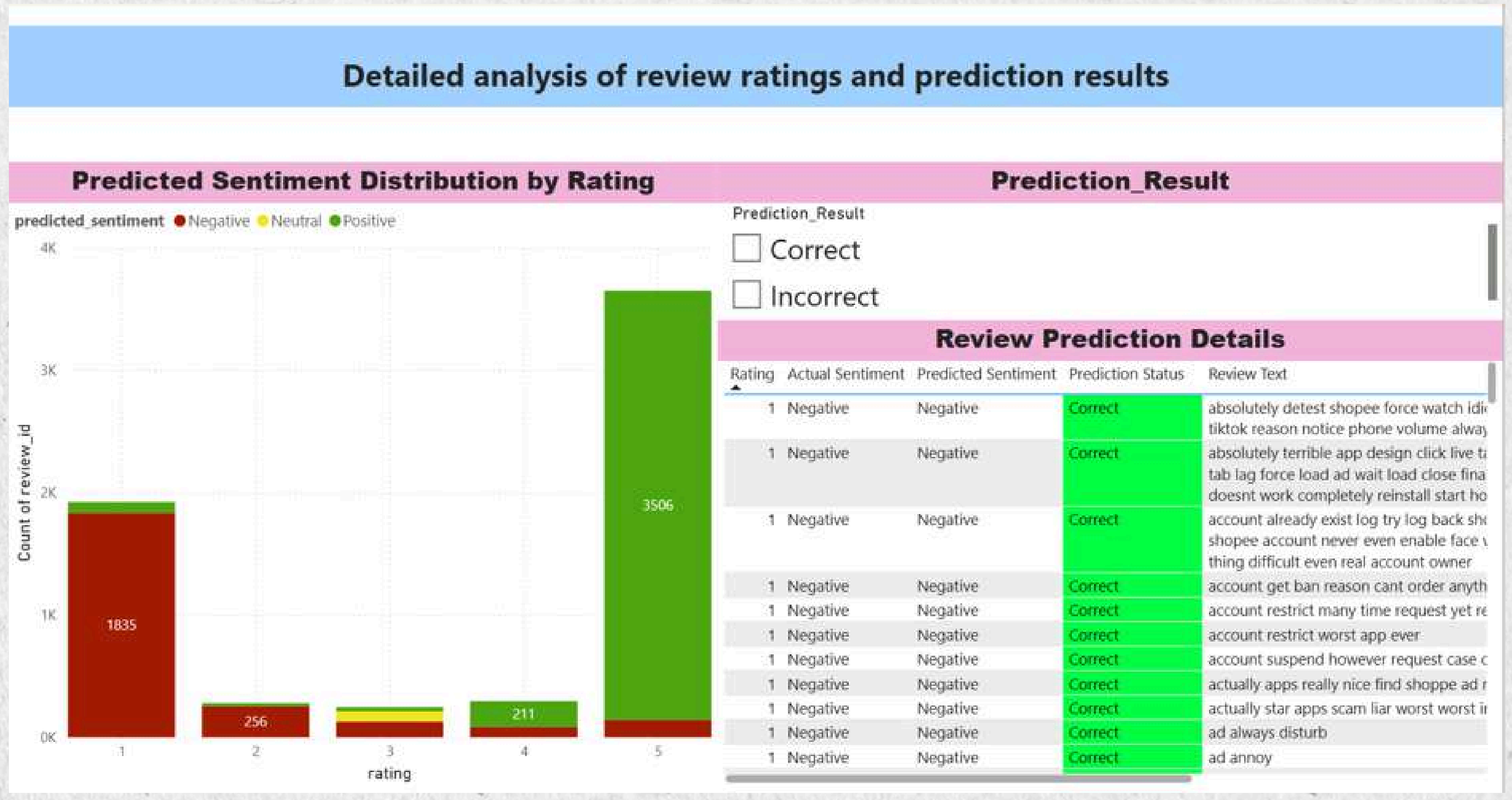
3. Output and Storage

- Saves results to streaming_prediction.csv
- Stores results in Elasticsearch
- Supports Kibana validation and Power BI dashboard

DASHBOARD

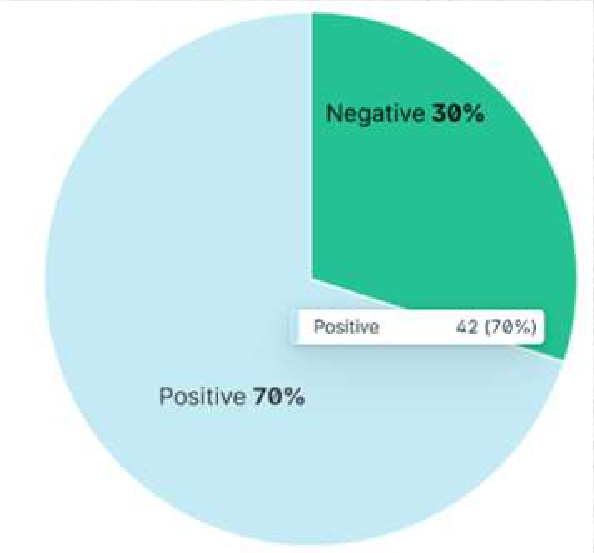
POWER BI DASHBOARD

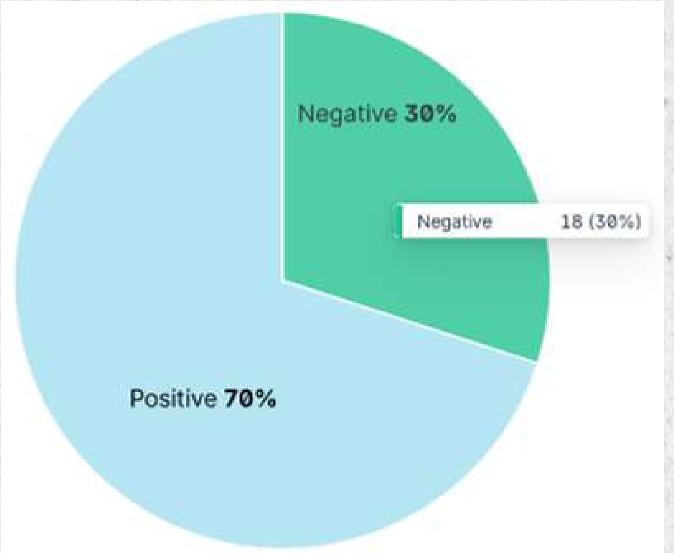




- 6395 reviews were analysed with 92.13% batch accuracy.
- Positive sentiment was dominant while neutral sentiment was the hardest to classify.
- Detailed review filtering helps identify correct and incorrect predictions.
- Both correct and incorrect details could be analyse at the second dashboard
- Rating 1,2 mostly are Negative reviews while rating 4,5 mostly are Positive review

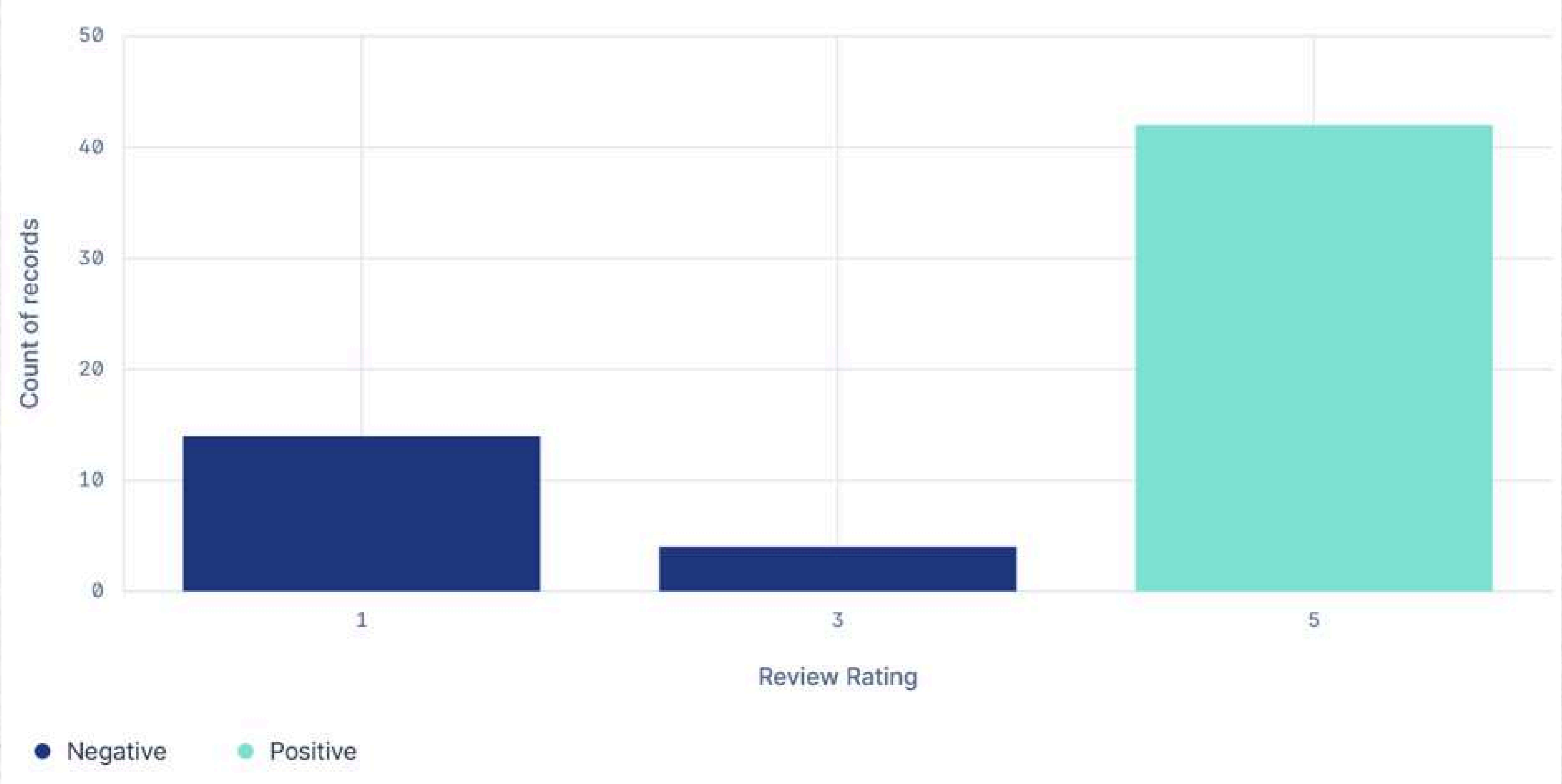
KIBANA DASHBOARD





Streaming Sentiment Distribution in Kibana

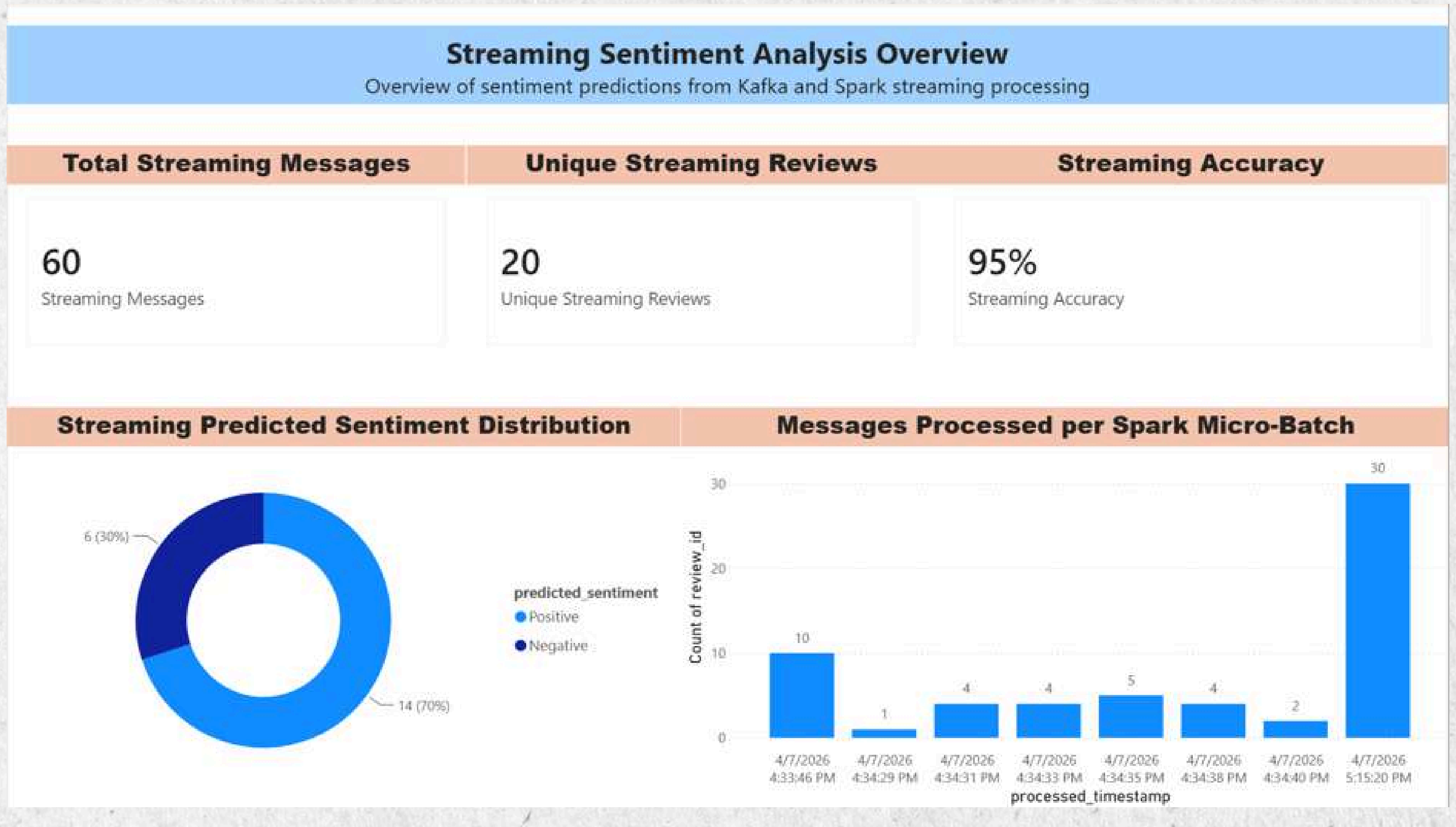
- 60 streaming messages were used and visualized in Kibana.
- Streaming sentiment consisted of 70% positive and 30% negative predictions.
- Due to small streaming samples, only positive and negative samples are selected.
- Spark processed the messages in multiple micro-batches over time.

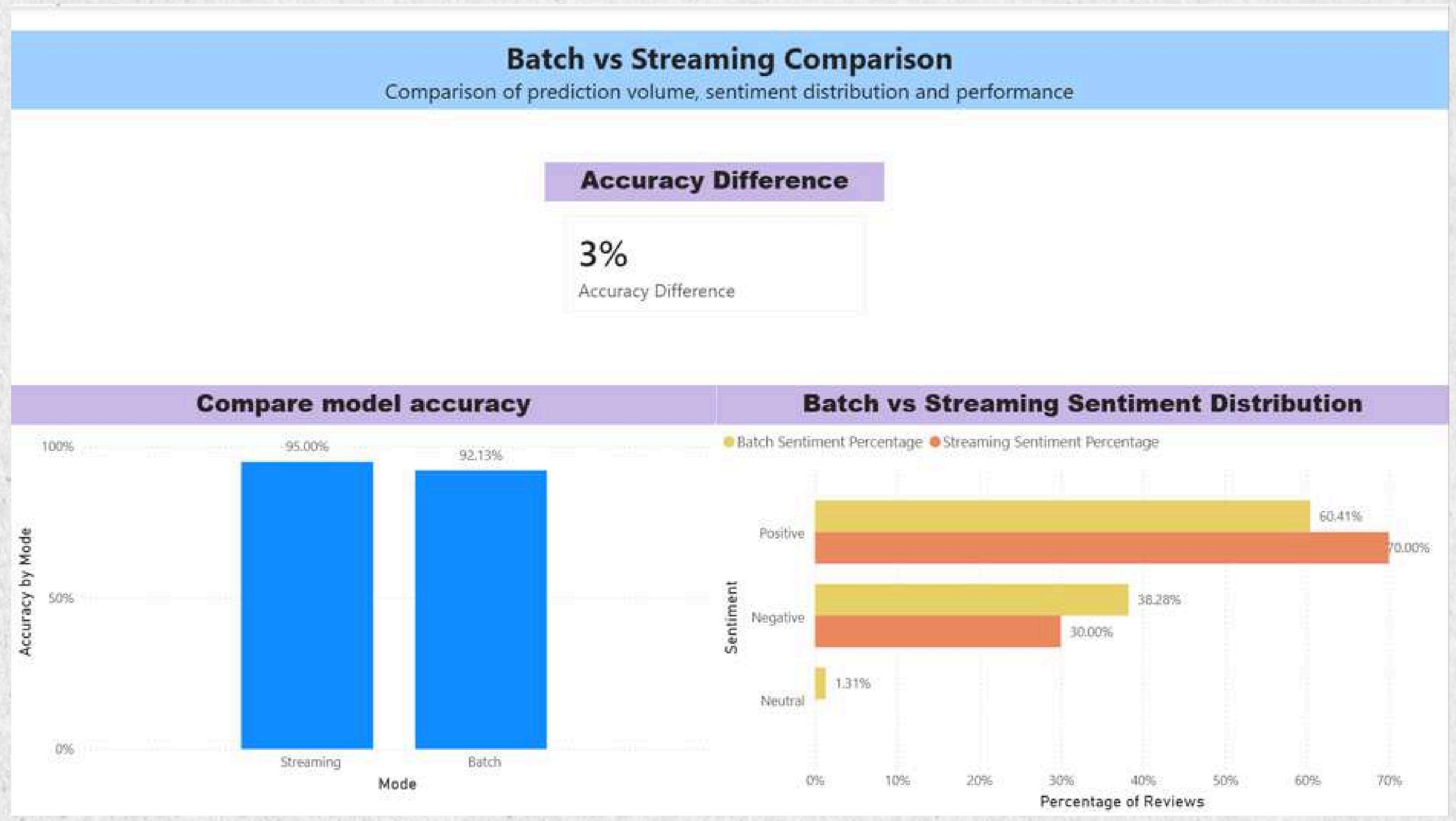


Predicted Sentiment by Review Rating

Sample Reviews Processed by Timestamp

POWER BI DASHBOARD





- There are total 60 streaming messages that represent 20 unique streaming reviews.
- Streaming achieved 95% accuracy compared with 92.13% for batch processing.
- Accuracy difference are only 3% difference
- Spark processed the messages in multiple micro-batches over time.
- Most of the reviews were positive and negative.
- Bigger streaming samples should be used for more accurate insights.

OPTIMISATION

OPTIMIZATION AND COMPARISON

Batch vs Streaming Comparison

- Batch: 6,395 reviews, 92.13% accuracy
- Streaming: 20/60 unique reviews, 95.00% accuracy
- Streaming result is higher but less reliable due to smaller sample size

Architecture Optimization

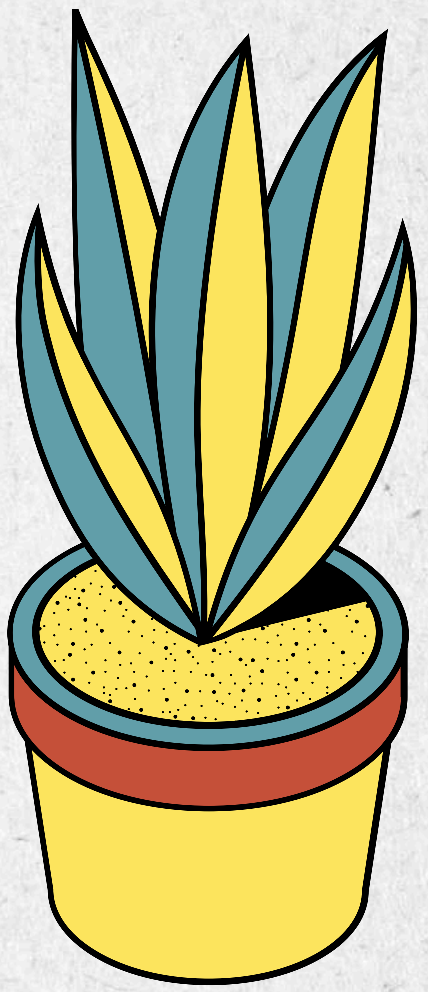
- Use larger streaming sample for fairer testing
- Increase Kafka partitions
- Tune Spark micro-batch interval and memory settings
- Add checkpointing, error handling and monitoring

Model Optimization

- Neutral sentiment was difficult to classify
- Collect more neutral reviews to balance the dataset
- Apply resampling or class-balancing techniques
- Try stronger models such as Logistic Regression, Linear SVM or BERT

Overall Discussion

- Batch is better for full model evaluation
- Streaming is better for real-time review processing
- Future work should improve model balance, scalability and performance monitoring



THANK YOU